Experiment 8: Evaluating and Registering Models using MLflow Registry

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AIM

To study model evaluation and registration using MLflow Model Registry and its integration with cloud-based registries.

OBJECTIVES

1. To understand model evaluation and version management in ML projects.
2. To study offline setup using local MLflow Registry with SQLite storage.
3. To understand online registration using Azure ML or Vertex AI Registry.

THEORY

1. Model Evaluation and Registry

After training, models must be evaluated and properly versioned before deployment.

MLflow provides a Model Registry to manage model versions, stages, and metadata. It allows comparison of different runs and controlled promotion of models to production, ensuring traceability and reliability.

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2. Offline Setup: Local MLflow Registry

In the offline setup, MLflow uses a local backend such as SQLite to store experiment details and model information.

Metrics can be analyzed using tools like Pandas, and models can be registered and versioned locally. This setup is suitable for development and testing.

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3. Online Setup: Cloud Model Registries

Microsoft Azure offers

Azure Machine Learning with integrated model registry features.

Similarly, Vertex AI provides a managed model registry.

These services support centralized version control, access management, and scalable deployment.

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4. Integrated Workflow

Models are first evaluated and registered locally using MLflow. After validation, selected versions are promoted to cloud registries for production deployment.

This ensures structured versioning, better governance, and scalable ML lifecycle management.

CONCLUSION

In this experiment, we studied evaluation and registration of ML models using MLflow locally and in cloud registries.

This approach improves version control, reproducibility, and reliable model deployment in real-world ML systems.